Predicting thermal performance of fired clay bricks lightened by adding organic matter - improvement of brick geometry

Laila Aouba[1,2,*], Marie Coutand[1], Bernard Perrin[1], Hervé Lemercier[2]

Abstract

Fired clay masonry bricks adapt to a variety of constraints during use. Through their mass, these products provide thermal inertia and thus a good indoor climate. Thanks to their thermal insulation properties, they help to minimize heating and cooling costs.
The brick industry allows different by-products to be added into the clay mixture at the beginning of the industrial cycle and the incorporation of organic matter improves thermal insulation while reducing the weight of the bricks.
This paper presents an experimental and numerical method for determining the thermal resistance of fired clay bricks containing residue of wheat straw (WS).
As their thermal conductivity reaches a value 35% lower than that of clay without organic matter, the thermal transmittance of such bricks can be improved by more than 20% relative to existing values.

Key words

Numerical simulation, hygrothermal modelling, fired clay bricks, organic matter, thermal conductivity

Introduction

In recent years, interest in green building materials and the valorisation of by-products from multiple industries has been increasing. As the brick industry allows a certain amount of organic and inorganic materials to be added during the mixing procedure, much research has been conducted to highlight the impact of these additions on the behaviour of fired clay bricks.
Waste material is used to replace a chosen amount of clay and the traditional method for producing clay bricks through fire kilns is followed. A number of reviews have reported the wide variety of waste that may be incorporated into bricks, including,

[1] Laboratoire Matériaux, Durabilité des Constructions, Université de Toulouse, FRANCE
[2] TERREAL, Route de Revel, 11400 Castelnaudary, FRANCE

[*] Corresponding author:
Laila Aouba, Laboratoire Matériaux, Durabilité des Constructions 135, avenue de Rangueil, F-31 077 Toulouse Cedex 04, FRANCE
Email: aouba@insa-toulouse.fr

among others, wood sawdust, paper production residue, Kraft pulp production residue, fly ash, cigarette butts, waste tea and rice husk ash (Ahmad, 2011; Bogdanov and al., 2012; Bories and al., 2014; Demir, 2006; Demir and al., 2005; Eliche-Quesada and al., 2012; Gökhan and Simsek, 2013; Kadir and al., 2009; Lingling and al., 2005; Sutcu and Akkurt, 2009; Tonnayopas and al., 2008). The literature also records several methods for measuring fired clay thermal conductivity, differences in the way the bricks are shaped (pressing or extrusion) and a range of firing temperatures, all of which make it difficult to compare the effect of pore-forming agents.

Nevertheless, the relationship between bulk density, porosity and thermal conductivity has been widely investigated and has shown that bulk density of bricks decreases when an amount of clay is replaced by organic matter, which leads to higher porosity and very low thermal conductivity of fired clay bodies (Banhidi and Gomze, 2007; García Ten and al., 2010; Muñoz and al., 2013; Phonphuak, 2013; Saiah and al., 2010; Sutcu and Akkurt, 2009). This correlation is illustrated in Figure 1; it also shows the large scatter due to very different manufacturing conditions as drying and firing temperature/duration or the type of kiln used for the production.

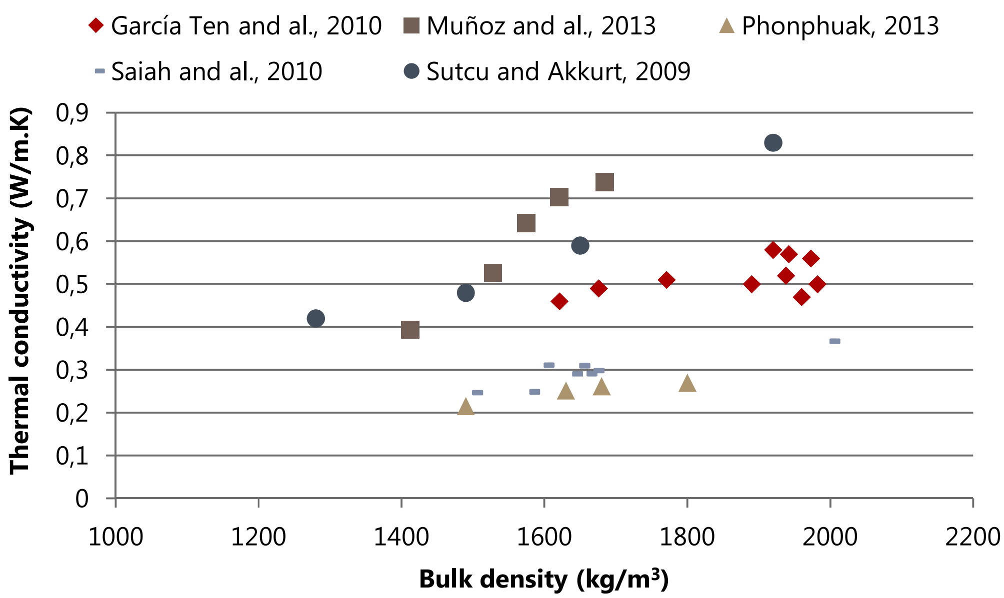

Figure 1 Thermal conductivity of clay bodies with pore-forming agents

The particle size also has a considerable effect on thermal conductivity. During firing, the added matter is consumed, leaving voids that increase the porosity.

The addressed question is: How is it possible to improve a formulation in terms of the percentages of included matter, sand and water and the firing temperature in order to obtain an industrial product with good thermal and mechanical characteristics. Considering fabrication cost of industrial products, it is necessary to work with laboratory samples but it should not be forgotten that their fabrication induces some anisotropies in the material properties. The idea is to identify the influence of different parameters on these properties in a first step and to validate the transition from the laboratory sample to the industrial product. Only partial results from a much larger work are presented in this paper, which describes the experimental and numerical methods for determining thermal resistance of fired clay bricks containing wheat straw residues (WS).

This paper will focus on the brick bulk density, thermal conductivity and geometry, the final purpose being to find optimized parameters that improve the properties of industrial products.

As the industrial products were too heavy and the cost of making them with the various research formulations was too high, the study was conducted on two types of products extruded in the laboratory but respecting industry requirements. The samples used were a cuboid block (170 mm x 75 mm x 17 mm) and laboratory alveolar bricks (78 mm x 78 mm x 80 mm). The cuboid block is used for physical and chemical characterisation whereas laboratory bricks are used for thermal characterisation.

Numerical calculations were performed on the laboratory alveolar bricks and on industrial-scale bricks (200 mm x 500 mm x 300 mm) to estimate thermal resistance on the basis of the measured thermal conductivities.

Properties of materials and samples

Materials

Clay was provided by one of the regional brick plants and the WS was obtained from a cooperative located in the south of France.

The chemical composition of the brick clay was determined by scanning electron microscopy (SEM) using a JEOL model JSM-6380 (high vacuum) microscope equipped for energy-dispersive X-ray analysis.

The mean results for three samples of bricks (C) are given in Table 1. The major elements were Si, Al and Fe.

Table 1 Chemical composition of the clay (% wt.)

	Al_2O_3	SiO_2	TiO_2	Fe_2O_3	K_2O	Na_2O	CaO	MgO	P_2O_5
Fired clay (C)	22.71	44.69	1.94	10.79	6.58	0.4	6.73	6.06	0.1
Std deviation	*±2.7*	*±2.9*	*±0.2*	*±0.4*	*±0.1*	*±0.1*	*±0.2*	*±0.2*	*±0.02*

The characteristics of WS were measured by LCA (Laboratoire de Chimie Agro-

Industrielle), a laboratory co-working in this project. The results are shown in Table 2.

Table 2 Composition of WS

	Dry matter	Proteins	Cellulose	Hemicellulose	Lignin	Lipids	Ash
WS (%)	90.86	2.18	42.12	11.88	28.92	4.57	8.89
Std deviation	*±0.07*	*±0.28*	*±1.07*	*±0.75*	*±0.88*	*±0.78*	*±0.22*

Various particle sizes of WS were incorporated into the clay mixture, the biggest being millimetrics.
The percentage of WS included in the mixture was between 4 and 8% by weight. The samples studied were identified as described below:

C (clay) - WS (wheat straw) - [G] (particle size)

- C- WS - very small (under 50 µm)
- C- WS – small (under 500 µm)
- C- WS – middle (over 500 µm)
- C- WS – large (Millimetric particle size)

Samples

The alveolar bricks and solid samples presented in Figure 2 were obtained by the same extrusion process for all formulations.

Figure 2 Fired clay brick samples: alveolar bricks (left) solid samples (right)

The samples were extruded at controlled pressure with an amount of water that gave the mixture sufficient plasticity for extrusion. The samples were then dried at 105°C before being fired at 920°C.
The process of fabrication induced an anisotropy (Fódi, 2011; Maillard and Aubert, 2014; Perrin and al., 2011) in the properties of the extruded material, as it created an orientation of the clay layers and consequently of the pore volume inside the material.

This particularity was taken into account in this work since it had an impact on the global behaviour of the industrial product.

Measurement of physical properties

The physical properties of open porosity, bulk density and thermal conductivity were determined for each formulation using samples extracted from the solid blocks.

Porosity and bulk density

Open porosity and bulk density were measured through a vacuum saturation test following the AFPC-AFREM (AFPC-AFREM Groupe de travail Durabilité des bétons 1998) recommendations. The bulk density $\rho\ (kg/m^3)$ and porosity $\varepsilon\ (\%)$ of the bricks were calculated from the following equations:

$$\rho = \frac{m_{dry}}{m_{air} - m_{water}} \times \rho_{water} \quad (1)$$

$$\varepsilon = \frac{m_{air} - m_{dry}}{m_{air} - m_{water}} \quad (2)$$

m_{dry} : oven-dry weight (kg)
m_{air} : saturated in air weight (kg)
m_{water} : saturated submerged weight (kg)
ρ_{water} : water density at 20 °C

Thermal conductivity

Thermal conductivity was determined according to the standardized methods ISO 8302 08-91, NF EN 1946-2 07-99 and NF X 10-021 12-72 using a hot plate apparatus (λ-Meter EP 500 from Lambda-Meßtechnik GmbH Dresden (Figure 3). Samples were prepared to fit to the equipment; the measurement area was 150 mm x 150 mm for a thickness of up to 10 mm. Thermal conductivity was obtained for 10° and 20°C with a 5°C temperature difference between the two plates. Four alveolar bricks were used, arranged as a block of 150 mm x 150 mm.

To account for the anisotropy, the measurements were carried out in the direction of the thickness for the blocks and at two positions for the alveolar bricks, as shown in Figure 3. The results obtained enabled the thermal conductivity in the parallel and perpendicular directions to be defined through the numerical simulation.

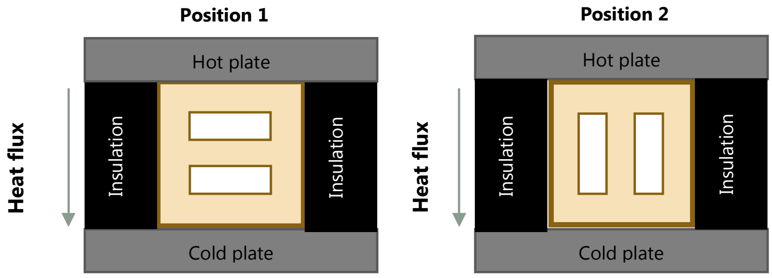

Figure 3 λ-Meter EP 500 measurements on alveolar laboratory bricks

The two-dimensional numerical heat transport simulation used to identify the conductivities of baked clay in the two directions relative to the extrusion direction was performed using DELPHIN 5, hygrothermal software developed by T. U. Dresden (Nicolai and Grunewald 2006). This numerical code simulates phenomena of coupled transport through porous media, such as energy, water, vapour, and air. In this work, only the energy balance was considered.

The aim of this simulation was to find the probable thermal conductivities of the parts parallel and perpendicular to the heat flux ($\lambda_{perp.}$ / $\lambda_{paral.}$) by comparing the simulated heat flux through the bricks with the flux measured experimentally.

For the simulation, the convective and radiative thermal transport effects of air in the holes are included in an equivalent empirical thermal conductivity coefficient according to the standard EN NF ISO 6946(NF EN ISO 6946 2008).

The purpose of this paper is not, actually, to develop physical phenomena involved within an air cavity, such as convective, radiative effects. The values of equivalent thermal conductivities given by the standard method take into account average values which can be considered as acceptable. These values have been largely experimented in the practice and works as measured values are compared with calculated ones.

Experimental and numerical results

Bulk density, porosity and thermal conductivity (values from tables of NF EN 1745)

Figure 4 shows the effect of adding organic materials to clay products. The parameters shown are porosity, dry bulk density obtained through a vacuum saturation test, and size of organic matter particles included.

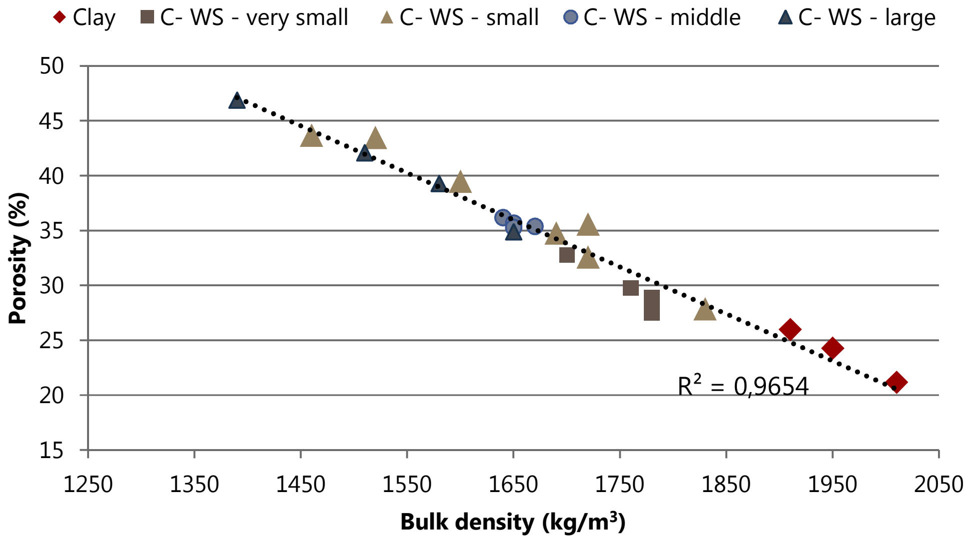

Figure 4 Evolution of porosity with bulk density

The study of this relationship will be presented in a future work.
The standard NF EN 1745 gives the procedure for obtaining the thermal performance of industrial products, based, in particular, on a very simple relation between bulk density of the baked clay material and its conductivity. This method does not take the anisotropy phenomena into account. In Figure 5, the expected conductivities for our materials are presented, depending only on their bulk density and according to NF EN 1745.

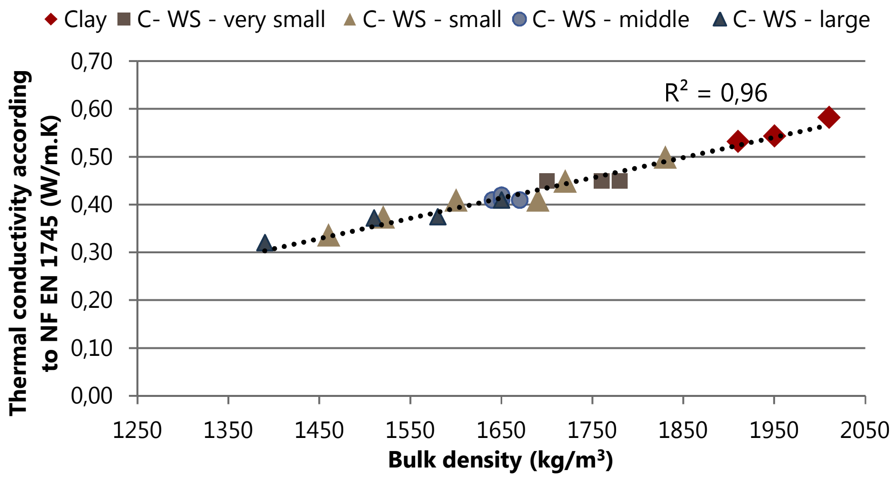

Figure 5 Evolution of thermal conductivity (NF-EN 1745) with bulk density

Thermal conductivity (Hot plate) and Numerical simulations by inverse method

To compare the thermal conductivity obtained following the standard NF EN 1745 (Annex A) with experimental values, some of the samples shown above were measured using the hot plate apparatus according to standardized methods ISO 8302 08-91, NF

EN 1946-2 07-99 and NF X 10-021 12-72. The hot-plate measurement was made at 20°C for two positions of the alveolar bricks as shown in Figure 3. The difference between the measurements at 10°C and 20°C was about 1 %.
The heat flux results and equivalent thermal conductivity given by this method are shown in Table 3.

Table 3 Hot plate apparatus results for vertical alveolar bricks

	Position 1			Position 2		
	Flux (W/m^2)	**λ (W/m.K)**	**R (m^2.K/W)**	**Flux (W/m^2)**	**λ (W/m.K)**	**R (m^2.K/W)**
C – Clay	22.1 ±1.1	0.35 ±0.023	**0.22**	25.2 ±1.2	0.37 ±0.021	**0.20**
C- 4%WS – very small	16.2 ±0.8	0.24 ±0.003	**0.31**	22.5 ±1.1	0.33 ±0.08	**0.23**
C- 4%WS - small	20.4 ±1.0	0.32 ±0.001	**0.24**	26.2 ±1.4	0.41 ±0.001	**0.19**
C- 4%WS - medium	18.1 ±0.9	0.28 ±0.004	**0.27**	19.4 ±1.0	0.30 ±0.02	**0.25**
C- 4%WS - large	14.2 ±0.7	0.21 ±0.006	**0.36**	17.4 ±0.9	0.26 ±0.001	**0.29**

From Table 3 it is apparent that Position 2 (heat flux parallel to hollows) gives higher flux values and lower thermal resistance values as position 1. The geometry of air cavity impacts this result.
To compare numerical response of DELPHIN 5 to experiment, an inverse method was used to simulated heat transfer for both positions (position 1 and 2) in the case of alveolar brick with clay formulation (C – clay). For each position, a couple of thermal conductivity ($\lambda_{paral.}$to heat flux and $\lambda_{perp.}$ to heat flux) was assigned to the clay material. Each thermal conductivity couple assigned gave a total heat flux for each position which was compared to experimental heat flux and thermal resistance.
The numerical simulations led to identify the most probable values for thermal conductivities depending on the direction of the flux. Frequency analyses gave an accurate solution for each value combining the two positions (Figure 6).

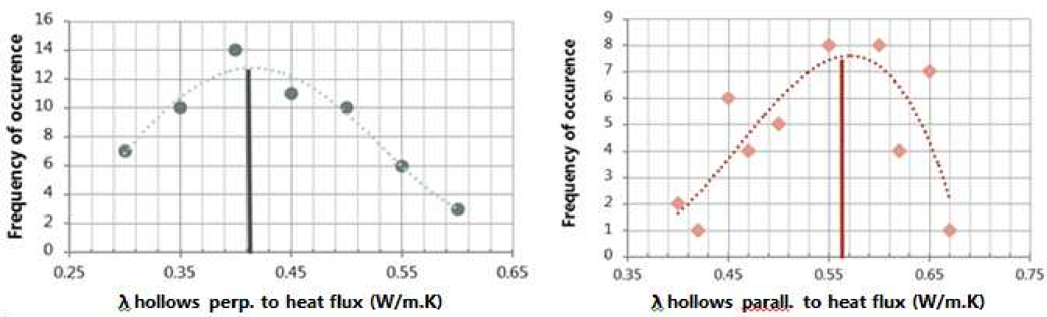

Figure 6 Analysis of the frequencies of probable conductivities

From the chart in Figure 7, we can see the correlation established between thermal conductivity and bulk density according to standard NF EN 1745, together with the values deduced from numerical simulation and the inverse method described above.

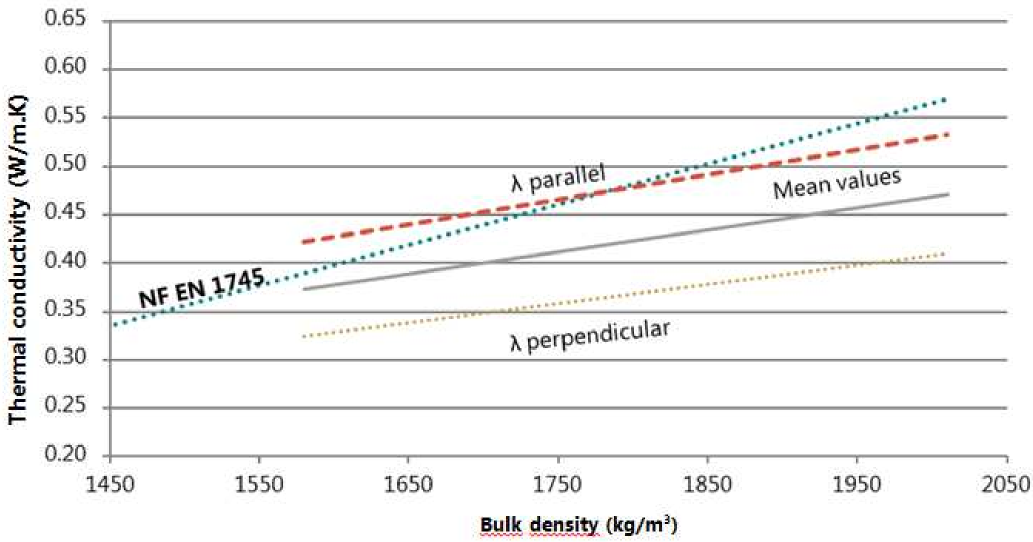

Figure 7 Relationship between thermal conductivity and bulk density

The most striking observation emerging from the data comparison is the difference between numerical and normalized slopes. The results from Annex A of NF EN 1745 do not consider the orientation of the porosity, unlike the values deduced from numerical simulation. As only WS is considered in this paper, more research on this topic needs to be undertaken to better understand the relations between the fabrication process, the orientation of porosity and the resulting thermal conductivities.

Computational simulations of porous clay bricks

In the light of the previous results and observations, the numerical code DELPHIN 5 (Nicolai and Grunewald 2006) is used to study the impact of thermal conductivity of the solid phase on the thermal resistance of industrial products in this section. Further work to find an optimum compromise between the thermal performance and mechanical behaviour of commercial products are in progress and will be presented in a later paper.
The different manufactured bricks presented in Figure 8 were considered in this study. The results highlight the effect of the thermal conductivity of clay bodies, and the effects of anisotropy and geometry.

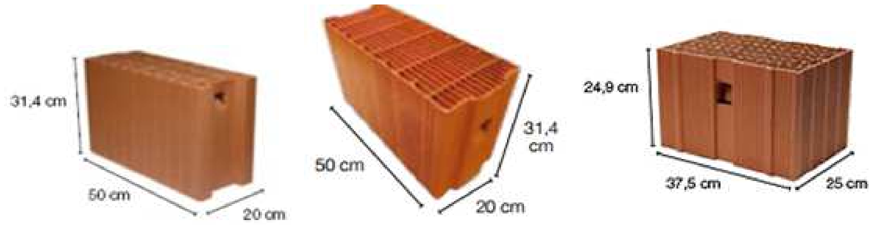

Figure 8 TERREAL Manufactured Products: BRICK 1, BRICK 2 and BRICK 3

The bricks presented in Figure 8 were meshed for the simulation of heat flux through the bricks. The boundary conditions are described in Table 4. The simulations are run for $\Delta\theta$ = 20 °C and humidity mass transfer is not considered in this case. Material inputs are assigned to air layers and solid phase (clay). The parameters considered for the materials are thermal conductivity and specific heat capacity.

Table 4 Boundary conditions

	Interior	Exterior
Temperature (°C)	25	5
Exchange coefficient for heat flow (W/m².K)	7.7	25

The first case of study was the impact of the thermal conductivity (considered isotropic for both directions: parallel and perpendicular to heat flow) on the thermal resistance for each type of industrial brick (Figure 9).

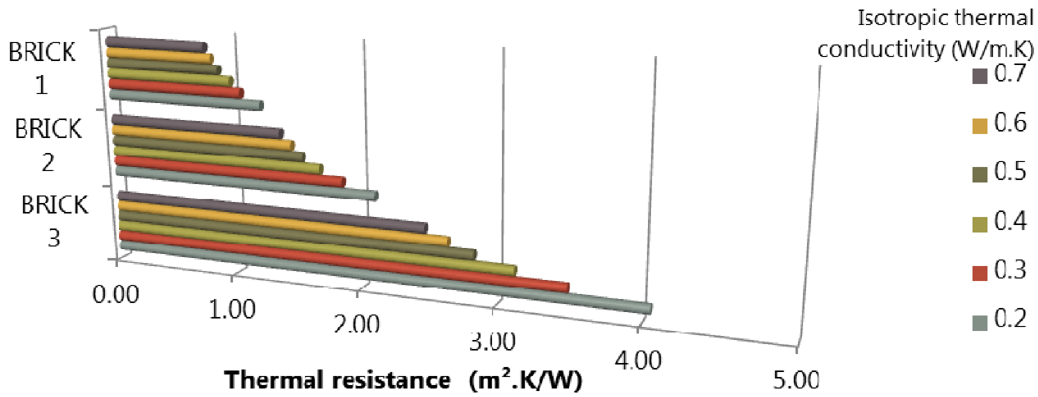

Figure 9 Evolution of thermal resistance for bricks 1, 2 and 3

The graph shows a slow increase of thermal resistance for brick 1 compared to brick 3. The effect of solid bodies with very low thermal conductivity (due to the incorporation of more organic matter) is more significant for bricks 2 and 3 where the geometry gains in

importance through the higher number of hollows and the sharp thickness of the partition walls. The gain is almost 2 m^2.K/W for brick 3 compared with a clay body with a thermal conductivity of 0.7 W/m.K.
It is interesting to note that the geometry combined with the use of porous clay bodies has a great effect in improving thermal resistance. The following simulations will highlight this effect by modifying the initial geometry.
Another parameter explored in Figure 10 is the anisotropy, which is taken into account in the simulation. The anisotropy ratio considered is 1.4 ($\lambda_{paral.}$ = 1.4 x $\lambda_{perp.}$). This ratio depends on the mean cross-sectional area of the alveolar bricks and the direction of the heat flow (Svoboda and Kubr 2010).

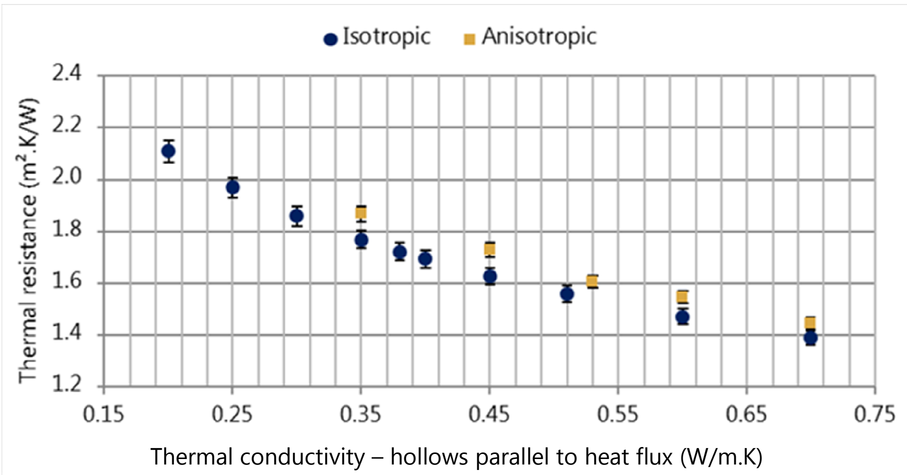

Figure 10 Thermal resistance vs. Thermal conductivity for partitions parallel to the heat flux

If the anisotropy is considered in the calculations, the thermal resistance obtained for the same thermal conductivity of the solid phase is higher than in the isotropic case.
The observed increase of $R_{th\text{-}anisotropic}$ could be attributed to the low values of $\lambda_{perpendicular}$ of the solid phase, taking the case of $\lambda_{parallel}$ = 0.35 W/m.K as an example, the equivalent $\lambda_{perpendicular}$ is around 0.25 W/m.K. As the number of partition walls is higher in the heat flow direction, the thermal resistance achieved is 6% higher than $R_{th\text{-}isotropic}$.

In the following section, the numerical simulation of thermal behaviour for Brick 2 has been realized for various partition and hollow thicknesses and for 3 thicknesses of bricks (200, 230 and 250 mm). The fixed parameters are the number of partition walls and hollows. Thermal conductivities $\lambda_{parallel}$, $\lambda_{perpendicular}$ are given in Table 5, boundary conditions are the same as in Table 4.
The purpose of this section is not to do a real optimisation of the geometry of the brick 2, but much more to analyse the impact of geometries lightly different from the geometry which has been defined and validated by the industrial producer.

Table 5 Brick 2: Parallel and perpendicular conductivities

Conductivity parallel to energy flux $\lambda_{paral.}$	Conductivity perpendicular to energy flux $\lambda_{perp.}$
0.60 W/m.K	0.47 W/m.K

The Figure 11 shows increasing thermal resistance values for thin walls and for larger bricks. To go beyond the existing thermal resistance value, it is interesting to maintain a hollow thickness greater than 9 mm with partition walls less than 7 mm thick.

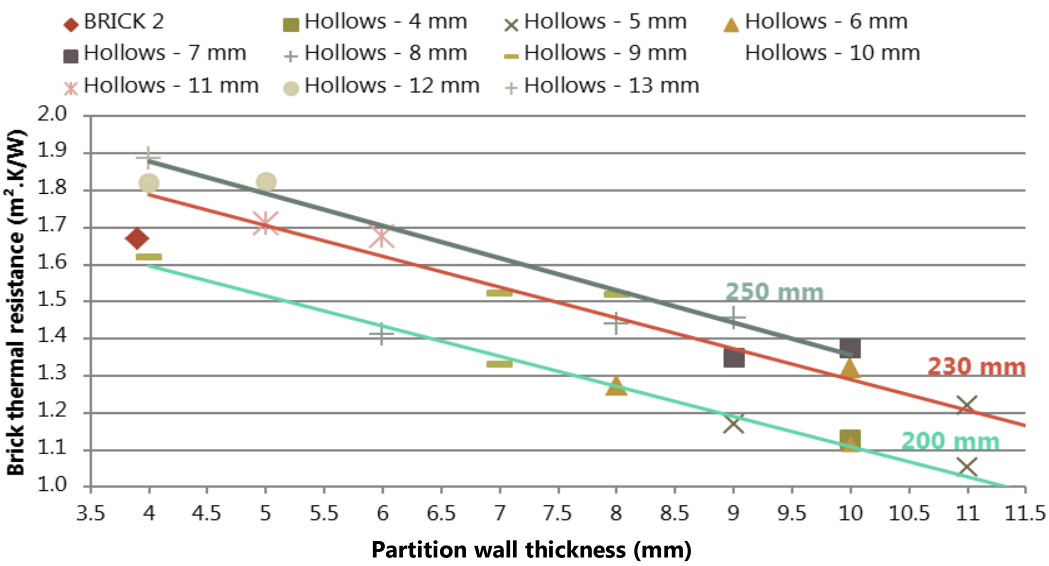

Figure 11 Thermal resistances for simulated bricks (vs. partition wall and hollow thicknesses)

As concluded in other works (M P Morales and al., 2011), the thermal transmittance of the bricks can be minimized by reducing thermal bridges and optimizing the arrangement of voids. It has to be considered that a smaller gap should be used when assembling bricks.

An optimized brick has to be easy to extrude; the thinner the partition walls are, the more difficult the brick is to produce as failures may occur in the solid phases.

These results provide further support for the brick industry. More parametric studies would be useful to find an optimum geometry that satisfies the requirements of the standards and is easy to process.

The following simulations (Figure 12) concern a brick thickness of 200 mm and a number of hollows fixed at 15. Thermal conductivity for walls perpendicular and parallel to the heat flux is modified from 0.60 to 0.39 W/m.K for $\lambda_{parallel}$ and from 0.47 to 0.30 W/m.K for $\lambda_{perpendicular}$.

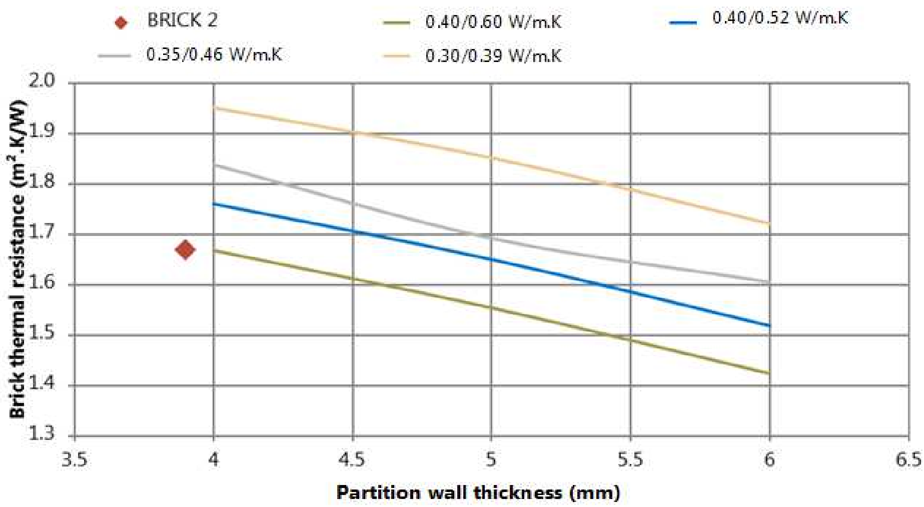

Figure 12 Thermal resistances for simulated bricks (vs. partition wall thickness and thermal conductivity)

It is interesting to note that the diminution of 35% in thermal conductivity values improves the brick's thermal resistance by around 20%. Increasing the thickness of partition walls has a negative impact on the thermal resistance of the bricks.

Taken together, these results suggest that there is an association between the thermal conductivity of the solid phases and the brick geometry. The most striking results are obtained for low thermal conductivity ($\leq$ 0.40 Wm.K for $\lambda_{parallel}$) and large numbers of hollows for bricks of 200 mm.

In this study, the vertical hollows were rectangular. More simulations could be done for different hollow geometries as seen in some works in the literature. Morales and al. (M.P. Morales and al., 2011) shows that geometries with partition walls that are not perpendicular to the heat flux reduce the wall's thermal conductivity. Also, thermal bridge in the tongue and groove can be avoided as the perforations of the voids are extended to tongue and groove area. According to literature, moreover reducing physical properties of fired clay bodies, decreasing the surface radiation emissivity is suitable to reduce thermal transmittance in bricks (Sutcu and al., 2014).

A part of brick 2 is shown in Figure 13, where the temperature field is displayed for configuration A (cf.Table 6).

Table 6 Properties of alveolar pattern

(W/m.K)	λ ∥	λ ⊥	Number of hollows
A	0.52	0.40	14
B	0.46	0.35	14
C	0.39	0.30	14
A1			12
A2	0.52	0.40	14
A3			15

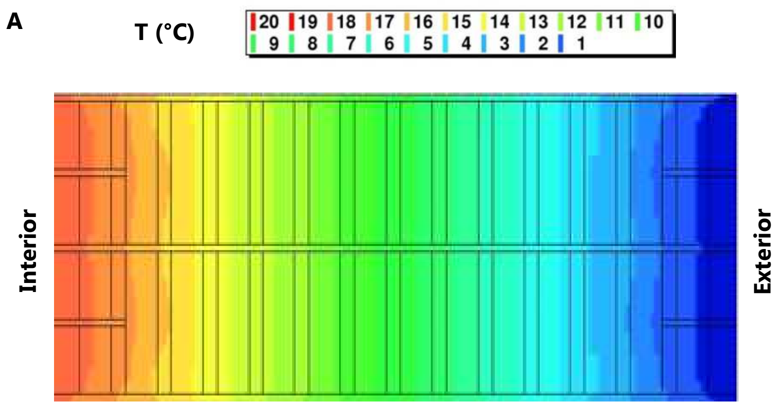

Figure 13 Temperature fields for case A

To improve the heat insulation, the geometry can be optimized as shown in Figure 14. The number of hollows across the brick width (200 mm) has been increased. Thermal conductivity values are: λ ∥ = 0.52 W/m.K and λ ⊥ = 0.40 W/m.K. A comparison between figures 14 and 15 shows a reduced expansion of the warm area, revealing a slight decrease in the thermal flux.

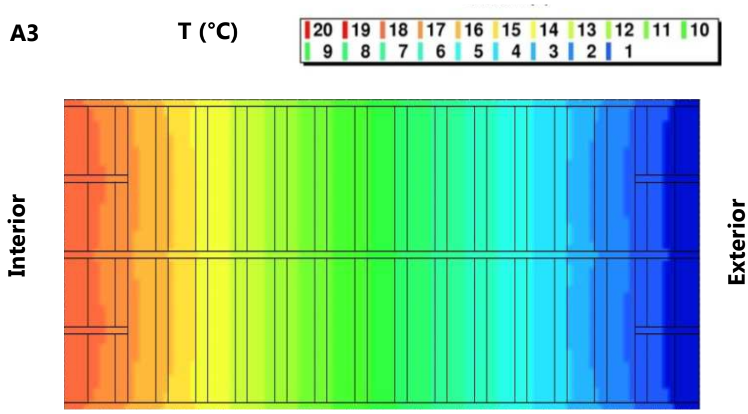

Figure 14 Temperature fields of 15-hollow simulations

The thermal resistances obtained for A1, A2 and A3 were respectively 1.59, 1.71 and 1.76 $m^2.K/W$. To keep a brick width of 200 mm, the thickness of the hollows had to be modified for a partition wall of 4 mm. For A1, hollows were 12 mm wide, while A2 and A3 had an alveolar size of 9 mm. This can explain the difference between their thermal resistances. The differences are due to the number of interfaces and also to the size of the hollows. As the size increases, other phenomena (such as radiation and air convection) become involved in heat transfer.

Conclusions

The main goal of the current study was to determine the influence of wheat straw incorporation and its effect on bulk density, porosity and clay body thermal conductivity. The correlation between bulk density and porosity and between bulk density and thermal conductivity was established. Decreased bulk density implies lower thermal conductivities, which permit industrial firms to produce lighter bricks (under 20 kg). Fired clay bricks are usually processed by extrusion, which causes anisotropy in the clay properties. This work shows that anisotropy affects thermal conductivity and it would be interesting to explore its effect on other properties such as gas permeability.
Considering the inverse method, the anisotropy of clay's thermal conductivity can be determined in relation to the heat flux and extrusion directions. Numerical simulations using DELPHIN 5 (Nicolai and Grunewald 2006) allowed the thermal transmittance gain of the bricks to be predicted by adding wheat straw to the clay mixture and improving the geometry of the bricks.

Numerical simulations of the three types of bricks have shown an impact of taking into account the anisotropy (through material assignments). The greater is the number of partition walls in the direction of heat flow the higher is the thermal resistance (+6%) compared to the R_{th} obtained for bricks with isotropic thermal conductivity assigned to the material in both direction (perpendicular and parallel to heat flow).
The diminution of the solid phase thermal conductivity (-35%) improve the brick's thermal resistance by around 20%. Furthermore the modification of the brick geometry (conserving 200 mm width) through the size of hollows and the number of interfaces may lead to better thermal resistance.

The current study has only examined thermal performance, but these findings should be considered in relation to the mechanical performance of the bricks, which are used for both their insulating properties and their load bearing capacity. Some numerical simulations are currently in progress to assess an optimized brick and to propose it at industrial scale.

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